ADEMP protocol: 04-treatment-main-effect

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This document is the working protocol for the exploratory simulation study that underlies `analysis/report/04-treatment-main-effect/` (full and short variants). The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019). Each study states its Aims (A), Data-generating mechanisms (D), Estimands (E), Methods (M), and Performance measures (P).

Common infrastructure

Sample sizes. Aggregated N-of-1 trial: $N = 35$ participants (per-design count, matching the Hendrickson 2020 minimum and the per-participant convention adopted in Blackston et al. 2019). Parallel-group RCT: $N = 35$ at 1:1 randomisation as the matched-N comparator, plus a sensitivity row at $N = 70$ (1:2 RCT-to-N-of-1 patient ratio).

Trial design (N-of-1 arm). Hendrickson hybrid design: 8 weeks open-label active drug + 4 weeks blinded discontinuation + 4 weeks crossover-1 + 4 weeks crossover-2, with assessment timepoints at the end of each period.

Trial design (RCT arm). 16-week parallel-group double-blind RCT with assessment at weeks 0, 4, 8, 12, 16.

Drug context. Prazosin for PTSD nightmares, calibrated to the Raskind 2013 PG-RCT and the @hendrickson2020 N-of-1 reference.

Carryover. Exponential decay with half-life $t_{1/2} = 3$ days at the baseline cell, varied across $t_{1/2} \in \{0.5, 1, 3, 5, 7\}$ days in the carryover sweep.

Replicates. $n_{\text{reps}} = 2000$ per cell, applied uniformly to power, bias, Type I error, and all sensitivity sweep cells. The target is MCSE $\leq 1.1\%$ on a power estimate near 0.75 and MCSE $\leq 0.5\%$ on a Type I estimate near 0.05; $n_{\text{reps}} = 2000$ satisfies both.

Random-number control. `RNGkind("L'Ecuyer-CMRG")` set once at program entry; `set.seed(2024L)` immediately after. Per- replicate `.Random.seed` state is captured before each replicate so that any anomalous result can be reproduced exactly. No `set.seed()` calls appear inside simulation functions.

Reporting standard. Morris, White and Crowther (2019) ADEMP with Monte Carlo standard errors for every reported performance measure.

Study M1 (primary). Power comparison at the prazosin baseline

A. Quantify the difference in statistical power for detecting the treatment main effect between the aggregated N-of-1 hybrid design and the matched parallel-group RCT, at the prazosin-PTSD-calibrated baseline parameters.

D. Mean-moderation Architecture A DGP with three response components (BR, PB, TV); true effect size -2.0 nightmares/week; carryover half-life 3 days; AR(1) within- participant residual correlation $\rho = 0.3$; between-participant random-intercept SD τ calibrated to yield $I^2 \approx 50\%$ in the parallel-group analysis.

E. Primary estimand: $\hat{\beta}_D$ in the analysis-model fixed-effects table (the average treatment effect coefficient). Secondary estimand: power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$.

M. N-of-1 arm: linear mixed-effects model `Sx ~ Dbc + t + (1 | ptID)` with `corCAR1(form = ~t|ptID)`. RCT arm: ANCOVA on endpoint with baseline severity covariate.

P. Power, type I error under $\beta_D = 0$, bias of $\hat{\beta}_D$, empirical SE of $\hat{\beta}_D$ across replicates, mean within-replicate model SE, nominal-95% CI coverage, convergence rate. Each reported with the binomial- proportion MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the Monte Carlo SE of the within-replicate mean estimator for continuous performance measures.

Studies S1-S12 (sensitivity sweeps)

Twelve sensitivity sweeps perturb one axis at a time around the M1 baseline. Each sweep fixes its own Aims, perturbed Data- generating mechanism axis, inherited Estimands, inherited Methods, and inherited Performance measures.

Sweep	Axis	Range	Cells
S1	Effect size β_D	$\{-3.0, -2.5, -2.0, -1.5, -1.0, -0.5, 0\}$	7
S2	Carryover $t_{1/2}$	$\{0.5, 1, 3, 5, 7\}$ days	5
S3	Decay-form mis-specification	linear / exponential / Weibull	3
S4	Between-patient SD τ	$\tau \in \{0, 0.5, 1, 1.5, 2\}$	5
S5	Within-patient SD σ	$\sigma \in \{0.5, 1, 1.5, 2\}$	4

Sweep	Axis	Range	Cells
S6	Cycles per patient	$k \in \{1, 2, 3, 4, 6\}$	5
S7	Total participant count	$N \in \{15, 25, 35, 50, 70, 100\}$	6
S8	AR(1) ρ	$\rho \in \{0, 0.15, 0.3, 0.45, 0.6\}$	5
S9	Period length	$L \in \{1, 2, 3, 4, 6\}$ weeks	5
S10	Biomarker interaction c_{bm} × DGP architecture	$\{0, 0.3, 0.6\} \times \{A, B\}$	6
S11	Carryover-by-AR(1) misspecification factorial	DGP $t_{1/2}$ × analysis $t_{1/2}$	16
S12	High-replicate null calibration	$\beta_D = 0$, $n_{\text{reps}} = 5000$	1

P (sensitivity-sweep specific). Each sweep reports power, bias, MCSE, and convergence rate at every cell. Results are plotted as a function of the perturbed axis with MCSE error bars.

Deliverables

- `analysis/scripts/treatment-main-effect/` driver scripts (already in place; documented in `analysis/scripts/treatment-main-effect/README.md`).
- ADEMP results table per sweep, with MCSE columns alongside every estimate.
- Figures: power-by-axis plot per sweep, plus the joint power-by-(N, effect-size) heatmap from M1.

Changes to the working protocol

Any change to the cells, estimands, methods, or performance measures specified in this working protocol is recorded in `02-deviations-log.md` with date, reason, and reviewer signoff.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/treatment-main-effect/00-ad`